

Pandrosion in Higher Dimensions, Part II

Complex Targets, Multi-Branch Convergence,
and Steffensen Acceleration via Rank-One Reduction

Abstract

This note extends the real Pandrosion dD iteration of the previous note [2] to targets $x \in \mathbb{C}$ and treats the acceleration problem. We prove three results. (i) The complex iteration retains exactly the same polynomial structure S_p^{dD} and admits the same algebraic fixed-point equations over $\mathbb{C}$, with multi-branch selection of the limiting root determined by the basin of attraction of the initial point. (ii) The rank-one Jacobian theorem of [2] survives unchanged in $\mathbb{C}$: any orbit lands on the diagonal $\{s_1 = \dots = s_n\}$ after exactly one iteration. (iii) As a consequence of (ii), the multivariate iteration reduces to a SCALAR fixed-point map after one step, so the classical Aitken Δ^2 (Steffensen) accelerator from paper 0 [1] applies directly to lift convergence from linear (rate λ_{dD}) to QUADRATIC. We do *not* need vector Brezinski-type accelerators: the rank-one collapse at $n = 1$ reduces the entire problem to one dimension in a single step. Numerical verification is at machine precision.

1 Setup

We follow the notation of [2]. Fix $p \geq 2$ and $d \geq 2$, set $n = d - 1$ and consider the multivariate geometric sum

$$S_p^{dD}(s_1, \dots, s_n) = \sum_{|\alpha| \leq p-1} s_1^{\alpha_1} \dots s_n^{\alpha_n}. \quad (1)$$

The dD Pandrosion iteration with target ratios $x_1, \dots, x_n \in \mathbb{C} \setminus \{0\}$ is

$$H_i(s) = 1 - \frac{x_i - 1}{x_i \cdot S_p^{dD}(s)}, \quad i = 1, \dots, n, \quad (2)$$

viewed as a holomorphic map $\mathbb{C}^n \rightarrow \mathbb{C}^n$. In the symmetric case $x_1 = \dots = x_n = x \in \mathbb{C}$, we write $\Delta_d(s) = S_p^{dD}(s, \dots, s) = \sum_{m=0}^{p-1} \binom{m+d-2}{d-2} s^m$.

2 Complex Pandrosion dD

2.1 Symmetric fixed points over $\mathbb{C}$

The polynomial fixed-point equation $x\Delta_d(s)(1-s) = x - 1$ defined in [2] continues to make sense for $x \in \mathbb{C}$ and specifies a polynomial of degree p in s . Hence the symmetric fixed-point set has at most p points in $\mathbb{C}$.

Theorem 1 (Complex fixed points). *For $p = 2$ and any $x \in \mathbb{C} \setminus \{0\}$, the equation $(d-1)xs^2 - (d-2)xs - 1 = 0$ has exactly two complex roots*

$$s_*^\pm = \frac{(d-2)x \pm \sqrt{(d-2)^2x^2 + 4(d-1)x}}{2(d-1)x}, \quad (3)$$

with $\sqrt{\cdot}$ the principal branch. For $p = 3$, the cubic $\binom{d}{2}xs^3 - \binom{d-1}{2}xs^2 - (d-2)xs - 1 = 0$ has exactly three complex roots.

Proof. Direct from the fundamental theorem of algebra applied to the same polynomial as in [2], but viewed over $\mathbb{C}$. The coefficients are rational in x , hence the roots are algebraic functions of x . $\square$ $\square$

2.2 Multi-branch convergence

For $x \in \mathbb{C}$ and $p \geq 2$, the iteration (2) has p symmetric fixed points (from Theorem 1). The orbit of H from a starting point s_0 converges to one of these fixed points (when convergence holds), and the choice depends on the basin of attraction.

Remark 2. For $x = 1 + i$ at $(d, p) = (3, 2)$, numerical experiments show convergence to $s_* \approx 0.847 - 0.209i$. Different starting points may land on the alternate root s_*^- ; the boundary between the two basins is the algebraic curve $\{s \in \mathbb{C} : \text{Im } s = \text{Im } H(s)\}$. We do not analyse this boundary here; the qualitative picture is that of paper 0 [1] extended to the complex plane.

2.3 Rank-one collapse persists in $\mathbb{C}$

Theorem 3 (Rank-one collapse, complex case). *In the symmetric case $x_1 = \dots = x_n = x \in \mathbb{C} \setminus \{0\}$, the map $H : \mathbb{C}^n \rightarrow \mathbb{C}^n$ from (2) sends every point $s \in \mathbb{C}^n$ in one iteration to the diagonal $\{s_1 = \dots = s_n\}$. That is, for any $s \in \mathbb{C}^n$, the components $H_1(s), \dots, H_n(s)$ are identically equal.*

Proof. By construction $H_i(s) = 1 - (x - 1)/(x S_p^{dD}(s))$ does not depend on i when all ratios x_i coincide. Hence all components of $H(s)$ are equal, lying on the diagonal of $\mathbb{C}^n$. $\square$ $\square$

This is a strictly algebraic statement, independent of the field; it holds verbatim for $\mathbb{R}$, $\mathbb{C}$, or any commutative ring containing the needed inverses. In particular, the spectral analysis of [2] (rank-1 Jacobian, eigenvalues $\{\lambda_{dD}, 0, \dots, 0\}$) carries over to $\mathbb{C}$.

3 Steffensen acceleration via rank-one reduction

The combination of Theorem 3 and the linear convergence rate λ_{dD} proved in [2] *automatically* reduces the multivariate accelerator design to the scalar one of paper 0.

3.1 The reduced scalar map

Define the scalar diagonal restriction of H :

$$g(s) := 1 - \frac{x - 1}{x \cdot \Delta_d(s)}, \quad s \in \mathbb{C}. \quad (4)$$

Lemma 4 (Reduction). *For any starting point $s_0 \in \mathbb{C}^n$, the orbit $s_0, H(s_0), H^2(s_0), \dots$ of (2) satisfies, for every $n \geq 1$, $s_n = (s_n^*, \dots, s_n^*)$ where s_n^* obeys the scalar iteration*

$$s_{n+1}^* = g(s_n^*).$$

Proof. By Theorem 3, after one iteration all components are equal: $s_1 = c\mathbf{1}$ for some $c \in \mathbb{C}$. On the diagonal $\{s_1 = \dots = s_n = s\}$, $S_p^{dD}(s, \dots, s) = \Delta_d(s)$ by definition, so $H_i(s\mathbf{1}) = 1 - (x - 1)/(x \Delta_d(s)) = g(s)$. Hence $H(s\mathbf{1}) = g(s)\mathbf{1}$, and the diagonal is invariant. Iterating, $s_n = s_n^*\mathbf{1}$ with $s_{n+1}^* = g(s_n^*)$. $\square$ $\square$

This reduction is the key. It says that for the symmetric problem, *only the scalar map g matters*, regardless of the dimension d : the multivariate aspect is entirely absorbed by the polynomial $\Delta_d(s)$. Acceleration techniques developed for scalar fixed-point iterations apply unchanged.

3.2 Aitken Δ^2 on the reduced map

The classical Aitken Δ^2 transformation [1, Section 12] applied to the scalar map g produces the Steffensen iterate

$$s_{n+1}^{\text{St}} = s_n - \frac{(g(s_n) - s_n)^2}{g(g(s_n)) - 2g(s_n) + s_n}. \quad (5)$$

This is a function of s_n requiring two evaluations of g (and hence two evaluations of Δ_d , i.e. four evaluations of S_p^{dD}).

Theorem 5 (Quadratic convergence of dD Steffensen). *Let s_* be a symmetric fixed point of g with $|g'(s_*)| = \lambda_{dD} < 1$ and $g'(s_*) \neq 1$. Then the Steffensen iterate (5), applied after the one-step diagonal projection of H , converges quadratically to s_* :*

$$|s_{n+1}^{\text{St}} - s_*| \leq K_S \cdot |s_n^{\text{St}} - s_*|^2 + O(|s_n^{\text{St}} - s_*|^3),$$

with $K_S = \frac{1}{2} |g''(s_*)| / |1 - g'(s_*)|$.

Proof. This is a direct application of the classical Steffensen theorem (see e.g. [1], Theorem 12.1) to the scalar map g at the fixed point s_* . All hypotheses are satisfied: g is holomorphic in a neighbourhood of s_* (rational with no pole there since $\Delta_d(s_*) \neq 0$); $g'(s_*) = \lambda_{dD}$ has modulus < 1 by [2]; $g'(s_*) \neq 1$ since $\lambda_{dD} < 1$. The constant K_S comes from the Taylor expansion of g at s_* . $\square$ $\square$

Remark 6 (Why no Brezinski-vector is needed). A naive multivariate generalisation would replace Aitken Δ^2 by Brezinski's vector Δ^2 (Wynn ε -algorithm at level 2), which uses the Samelson inverse of vectors. Such an extension is unnecessary here: the rank-one collapse of Theorem 3 reduces the orbit to a one-dimensional sequence in $\mathbb{C}$ after a single iteration, on which the scalar Aitken Δ^2 already provides quadratic order. This is a remarkable feature of Pandrosion's geometric symmetry: the multivariate "coupling" through S_p^{dD} is so strong that all orbits collapse to one dimension.

4 Numerical verification

The companion script `flow/005_pandrosion_dD_complex_steffensen.py` verifies each theorem to machine precision. Highlights:

- **Theorem 1:** orbits with various $x \in \mathbb{C}$ (e.g. $x = 1 + i$, $2 + 0.5i$, $1.5 + 2i$) at $d \in \{3, 4, 5\}$ and $p \in \{2, 3\}$ converge to the corresponding algebraic fixed points with residual $|x\Delta_d(s_*)(1 - s_*) - (x - 1)| < 10^{-13}$.
- **Theorem 3:** starting from $s_0 = (0.1+0.2i, 0.4-0.1i, 0.6+0.3i, 0.9-0.5i)$ at $(d, p) = (5, 2)$, $x = 1 + i$, the orbit lands on the diagonal at step $n = 1$ with component-spread exactly 0.
- **Theorem 5:** at $(x, d, p) = (1.5 + 0.5i, 4, 2)$, the orbit error sequences are

Linear: $6.2 \cdot 10^{-2}, 6.4 \cdot 10^{-3}, 6.3 \cdot 10^{-4}, 6.3 \cdot 10^{-5}, \dots$

Steffensen: $6.2 \cdot 10^{-2}, 3.6 \cdot 10^{-5}, 1.2 \cdot 10^{-11}, 5 \cdot 10^{-15}$.

Empirical convergence orders measured by the formula $\log(\varepsilon_{n+1}/\varepsilon_n) / \log(\varepsilon_n/\varepsilon_{n-1})$ are 1.02 (linear) and 2.02 (Steffensen), matching the theoretical 1 and 2. Steffensen converges to machine precision in 4 iterations, vs. 14–17 for the raw dD iteration.

- **Multi-branch:** at $x = 2$, $d = 3$, $p = 3$, the cubic fixed-point equation has one real root and two complex conjugate roots; varying s_0 produces convergence to the real root in this test, but starting points further into the complex plane reach the conjugate roots.

5 Conclusion

The complex extension of Pandrosion dD is straightforward: the entire algebraic and dynamical structure of [2] carries over to $\mathbb{C}$. The unexpected payoff is the acceleration analysis. Because the rank-one Jacobian projects every orbit onto the diagonal in a single step, the multivariate iteration is *indistinguishable from a scalar one after one iteration*. Steffensen’s classical accelerator from paper 0 thus applies directly and produces quadratic convergence in arbitrary dimension $d \geq 3$, without invoking vector Padé-Brezinski-Wynn machinery.

This decoupling between the multivariate algebraic structure (encoded by S_p^{dD}) and the temporal acceleration structure (Aitken Δ^2 on the reduced scalar map) is the central simplifying phenomenon of Pandrosion in arbitrary dimensions. In particular, the quadratic-convergent algorithm

1. Run one iteration of H to project to the diagonal.
2. Apply scalar Steffensen Δ^2 on the reduced map g until desired accuracy.

yields a d -dimensional Pandrosion solver of the same asymptotic cost as the 2D one of paper 0, while inheriting the stronger contraction rate $\lambda_{dD} \sim 1/(2d)$ before acceleration.

References

- [1] A formally verified Pandrosion–Steffensen scheme, `0pandrosion_pth.tex`.
- [2] Pandrosion in higher dimensions: simplex-lattice generalisation, closed-form contraction, and a rank-one spectral theorem, `1pandrosion_dD.tex`.